

SHREEHARI AVADHANI

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Summary

CSE student with strong foundations in Python, C++, SQL, data structures and machine learning. Experienced in building end-to-end AI systems and backend applications using Python, FastAPI and PostgreSQL, with hands-on experience developing, evaluating and deploying AI/ML prototypes. Strong interest in applying AI and machine learning to real-world problems.

Technical Skills

Languages: Python, C++, SQL

AI/ML: Machine Learning, LLMs, RAG, Embeddings, Sentence Transformers, LangChain, LangGraph

Backend/Tools: FastAPI, REST APIs, Docker, Git, GitHub

Databases: PostgreSQL, pgvector, Redis, Supabase

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Education

Nitte Meenakshi Institute of Technology

B.E. Computer Science and Engineering

Oct 2023 – July 2027

Bengaluru, KA

- CGPA: 8.05

Projects

NMIT-GPT – Institutional AI Chatbot | 🤖 | *Python, FastAPI, PostgreSQL, pgvector, RAG* **2026**

- Built an end-to-end AI assistant using Python and FastAPI for institutional academic queries, integrating PostgreSQL, pgvector, Sentence Transformers, and Groq LLMs.
- Implemented semantic vector retrieval, SQL-based structured querying, and LLM-based intent detection to route structured and unstructured requests.
- Implemented multi-turn session handling and pronoun resolution to maintain context across follow-up queries.

Multitenant RAG System with Semantic Caching | 🤖 | *Python, FastAPI, PostgreSQL, pgvector, Redis, Docker* **2026**

- Built and evaluated a production-oriented multitenant AI retrieval system using Python, PostgreSQL and pgvector, with tenant isolation enforced at the SQL layer.
- Implemented hybrid retrieval using BM25 and pgvector dense search with Reciprocal Rank Fusion and cross-encoder reranking to improve document relevance.
- Evaluated retrieval and response quality using RAGAS, achieving **0.75 Faithfulness, 0.75 Answer Relevancy, 0.83 Context Precision**, and **0.73 Context Recall**.
- Containerized the system using Docker Compose and implemented Redis semantic caching, IVFFlat vector indexing, health checks, and auto-recovery.

Intelligent Research Agent | 🤖 | *Python, LangGraph, FastAPI, PostgreSQL, pgvector, Docker* **2026**

- Explored Agentic AI by building a multi-tool AI prototype using Python and LangGraph, where an LLM selects tools to answer user queries.
- Integrated knowledge-base retrieval, web search, and mathematical computation as tools, connecting the agent to the Multitenant RAG system through a REST API.
- Added conversational memory using PostgreSQL, pgvector, and Sentence Transformers.
- Exposed the application through FastAPI and Docker Compose.

Leadership

Esports Co-Lead, Oasis (NMIT Events) | Coordinated institute-level esports tournaments, managing registrations, scheduling, brackets, match execution, and participant communication.